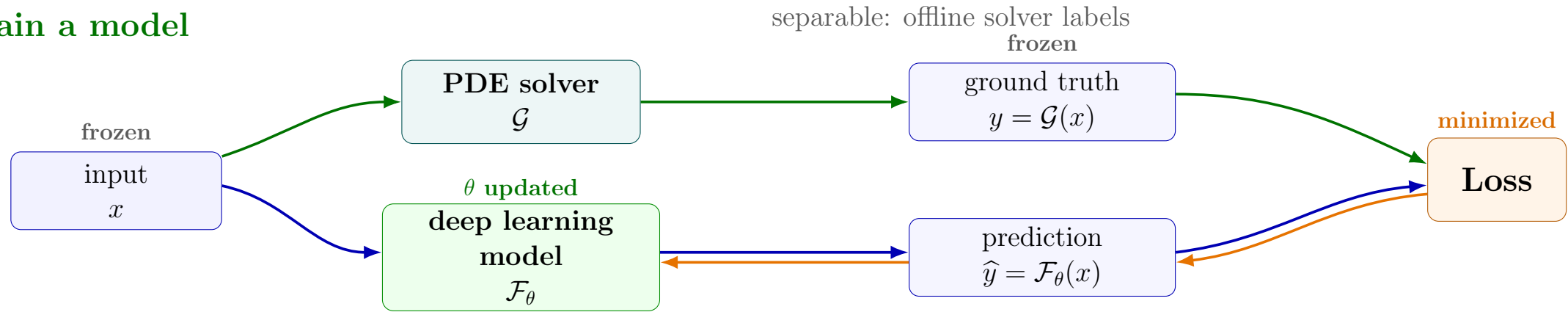


supervised model training and adversarial attacking in the context of PDE operator learning

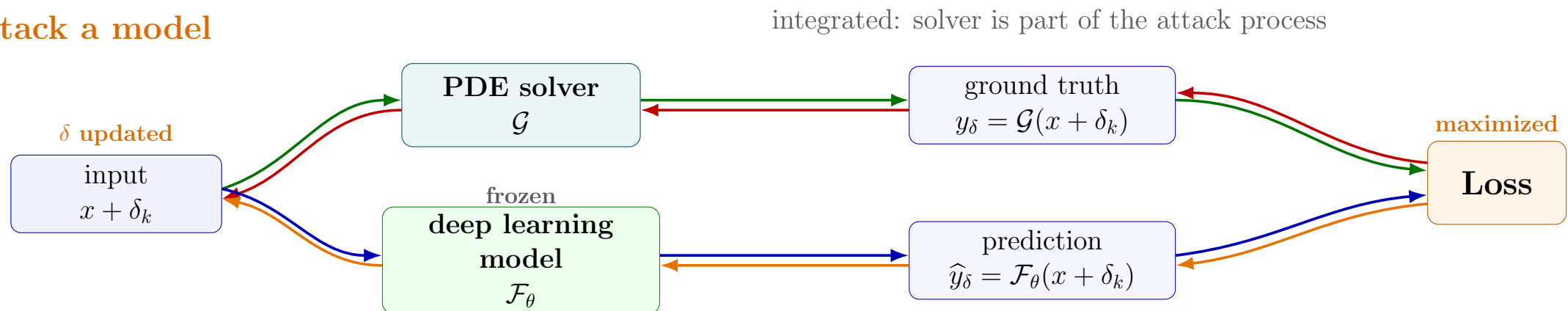
→ model forward
 → solver forward
 → model backward
 → solver backward

Train a model



$$\mathcal{L}_{\text{train}}(\theta; x) = \frac{1}{2} \|\mathcal{F}_{\theta}(x) - \mathcal{G}(x)\|_2^2; \quad \nabla_{\theta} \mathcal{L}_{\text{train}} = D_{\theta} \mathcal{F}_{\theta}(x)^* (\mathcal{F}_{\theta}(x) - \mathcal{G}(x)); \quad \theta_{t+1} = \theta_t - \alpha \nabla_{\theta} \mathcal{L}_{\text{train}}(\theta_t; x).$$

Attack a model



$$x_k = x + \delta_k, \theta \text{ fixed}; \quad \mathcal{L}_{\text{adv}}(\delta_k) = \frac{1}{2} \|\mathcal{F}_{\theta}(x_k) - \mathcal{G}(x_k)\|_2^2; \quad \nabla_{\delta} \mathcal{L}_{\text{adv}} = (D_x \mathcal{F}_{\theta}[x_k] - D_x \mathcal{G}[x_k])^* (\mathcal{F}_{\theta}(x_k) - \mathcal{G}(x_k)); \quad \delta_{k+1} = \Pi_{\Delta}(\delta_k + \eta \nabla_{\delta} \mathcal{L}_{\text{adv}}(\delta_k)).$$